

AI-POWERED MEDICAL DOCUMENTATION SYSTEM

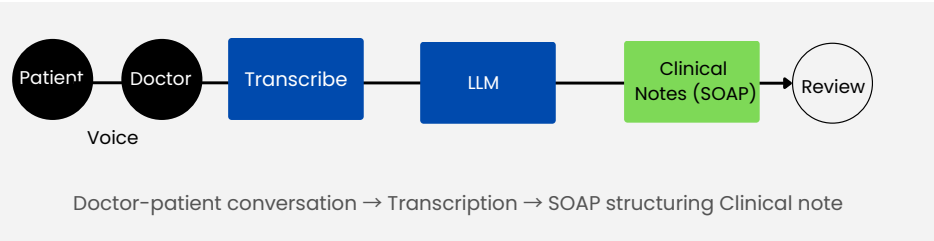
Re-inventing Patient Care with AI Workflow Automation

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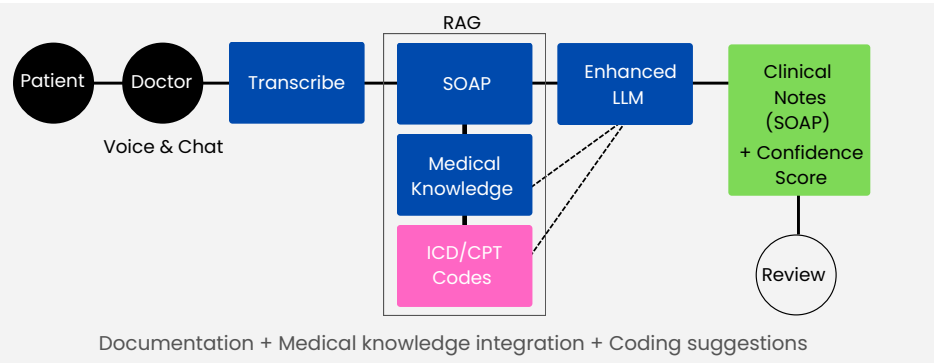
USE CASE

Building a progressive AI-Powered medical documentation system that evolves from basic transcription to intelligent workflow automation, **ultimately reducing physician burn-out and improving patient care.**

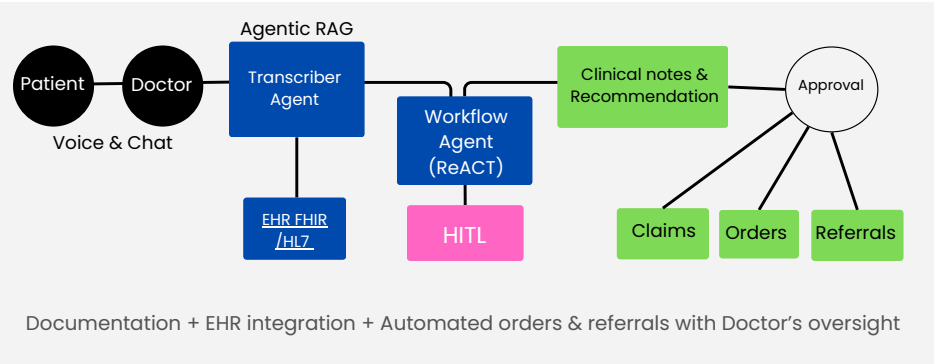
ITERATION 1: Med-Transcriber (Basic - Voice to text)



ITERATION 2: Med-Assistant (RAG)



ITERATION 3: Med-Partner (Agentic RAG + ReACT). - MEDCARE



SYSTEM DESIGN

Business Constraints:

- Real-time processing with minimal latency for clinical efficiency
- Specialized models** will be leveraged, trade-off between models will be **finalized after 1st iteration** required for advanced clinical reasoning capabilities in iteration 3.
- HIPAA compliance and FDA SaMD regulatory framework mandatory
- HITL review + model observability**

Iteration 1: Med-Transcriber (Testing)

- Leveraging tools like Whisper to convert patient-doctor audio conversations to text
- LLM analyses and classifies text into SOAP format as clinical note to physician
- Audio-to-text conversion → LLM Summary into SOAP structure → Review

Iteration 2: Med-Assistant

- Maintains audio transcription process from Iteration 1 → DB
- Retrieves existing patient medical data (RAG) → LLM → Clinical Notes
- Enhances clinical notes grounded with Clinical Ontologies
- Suggest accurate ICD-10/CPT codes with confidence scores to the physician

Iteration 3: Med-Partner (2 phase deployment)

- RAG system enhanced with EHR FHIR integration and agentic workflow
- Document Agent analyzes clinical documentation to identify required medical actions
- Workflow Agent processes orders, referrals, and follow-ups pending physician approval
- HITL maintains mandatory physician review and sign-off with attestation tracking

Iter.	Cost/Latency Factors	Optimizations	Guardrails	Eval Metrics
1	Whisper + LLM call	Medical prompt templates	PHI filtering	SOAP completeness, transcription accuracy
2	+ Vector DB retrieval, LLM Calls	Hybrid search + metadata	Document access control	ICD/CPT accuracy, hallucination rate
3	+ LLM & API calls to EHR, order systems	ReAct, short-term memory	HITL approval workflow	Task success, tool accuracy, time saved